10.16 (a) The differential cross section, summed over outgoing polarization, is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{E_0^2} \sum_\lambda \left| \vec{\varepsilon}_\lambda^{\,*} \cdot \vec{F}_{sh} \right|^2 = \frac{k^2}{4\pi^2} \int_{sh} d^2x_\perp \int d^2x_\perp' \exp\{ i \vec{k}_\perp \cdot (\vec{x}_\perp - \vec{x}_\perp') \} \sum_\lambda \left| \vec{\varepsilon}_\lambda^{\,*} \cdot \vec{\varepsilon}_0 \right|.$$

Adopting the Einstein summation convention, and noticing that

$$\sum_\lambda \varepsilon_{\lambda,i}^{*} \, \varepsilon_{\lambda,j} = \delta_{ij} - \hat{n}_i \hat{n}_j,$$

where $\hat{n}$ is the unit vector in the direction of $\vec{k}$, the summation becomes

$$\sum_\lambda \left| \vec{\varepsilon}_\lambda^{\,*} \cdot \vec{\varepsilon}_0 \right| = \sum_\lambda \varepsilon_{\lambda,i}^{*} \varepsilon_{0,i} \varepsilon_{\lambda,j} \varepsilon_{0,j}^{*} = (\delta_{ij} - \hat{n}_i \hat{n}_j) \varepsilon_{0,i} \varepsilon_{0,j}^{*} = 1 - |\vec{\varepsilon}_0 \cdot \hat{n}|^2.$$

In the short-wavelength limit $\vec{k} \simeq \vec{k}_0$, and $\vec{\varepsilon}_0^{\,*} \cdot \vec{k} \simeq 0$. Therefore,

$$\frac{d\sigma}{d\Omega} = \frac{k^2}{4\pi^2} \int_{sh} d^2x_\perp \int_{sh} d^2x_\perp' \exp\{ i \vec{k}_\perp \cdot (\vec{x}_\perp - \vec{x}_\perp') \},$$

and

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = \frac{1}{4\pi^2} \int_{sh} d^2x_\perp \int_{sh} d^2x_\perp' \int d^2k_\perp \exp\{ i \vec{k}_\perp \cdot (\vec{x}_\perp - \vec{x}_\perp') \}$$

$$= \int_{sh} d^2x_\perp \int_{sh} d^2x_\perp' \, \delta(\vec{x}_\perp - \vec{x}_\perp') = \int_{sh} d^2x_\perp = S_{sh}$$

(b) In the forward direction, $\vec{k} = \vec{k}_0$, $\vec{\varepsilon} = \vec{\varepsilon}_0$, $\vec{k}_\perp = 0$, and

$$\vec{\varepsilon}_0^{\,*} \cdot \vec{f}(\vec{k} = \vec{k}_0) = \frac{\vec{\varepsilon}_0^{\,*} \cdot \vec{F}(\vec{k} = \vec{k}_0)}{E_0} = \frac{ik}{2\pi} S_{sh}$$

By optical theorem, the total scattering cross section is

$$\sigma_t = \frac{4\pi}{k} \operatorname{Im}\left[\vec{\varepsilon}_0^{\,*} \cdot \vec{f}(\vec{k} = \vec{k}_0) \right] = \frac{4\pi}{k} \cdot \frac{k}{2\pi} S_{sh} = 2 S_{sh},$$

i.e., twice of the shadow area.